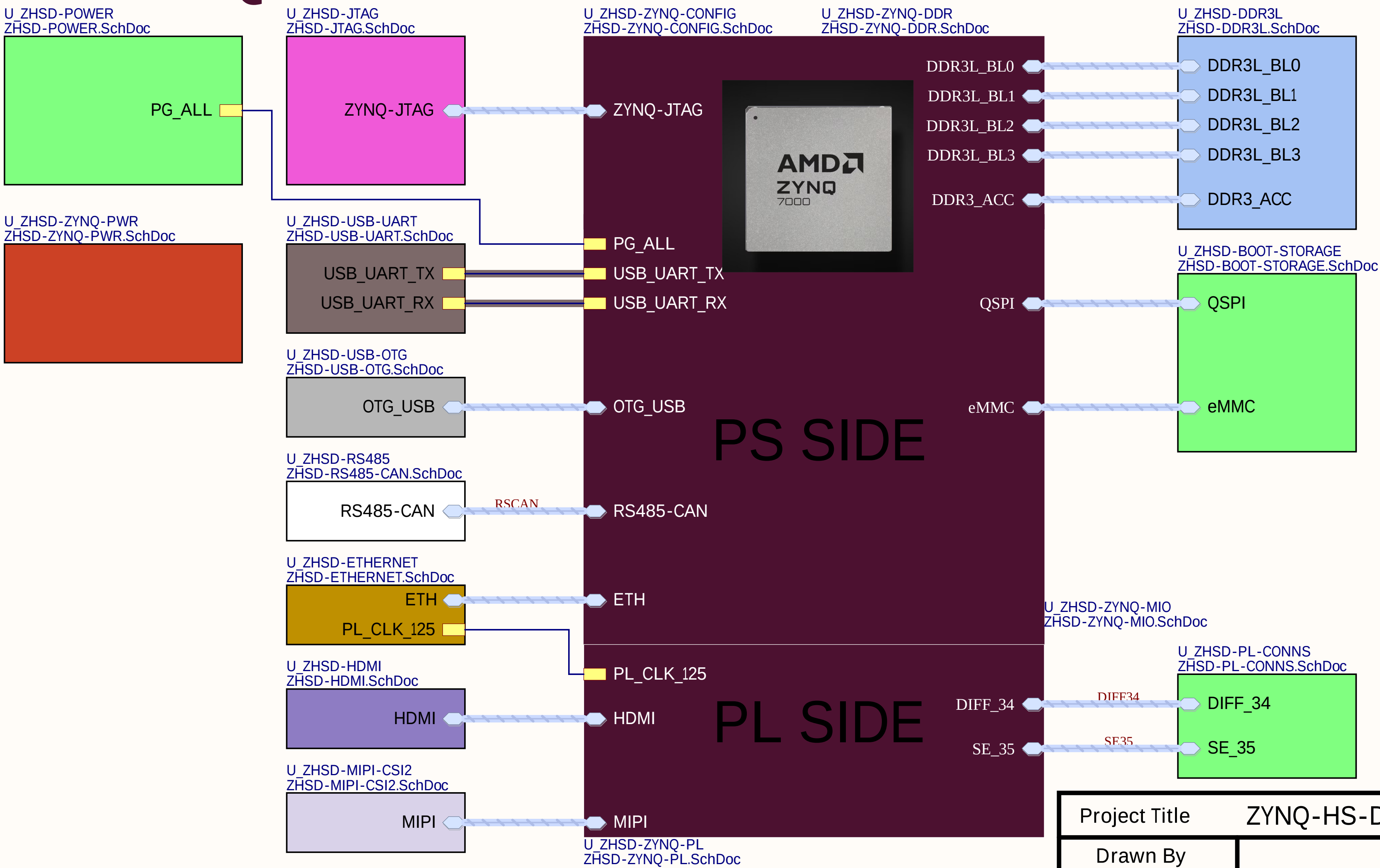
ZYNQ-7000 HIGH SPEED DESIGN



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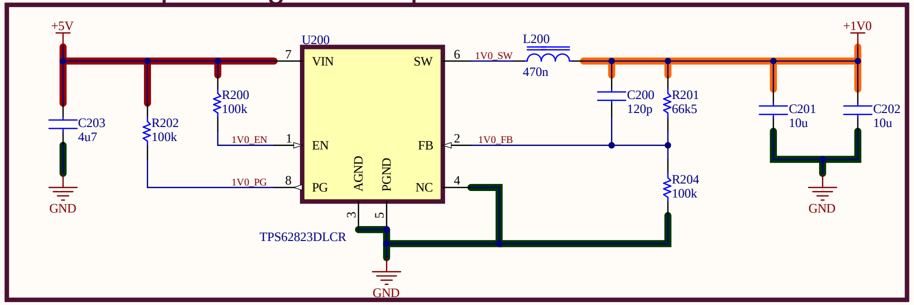
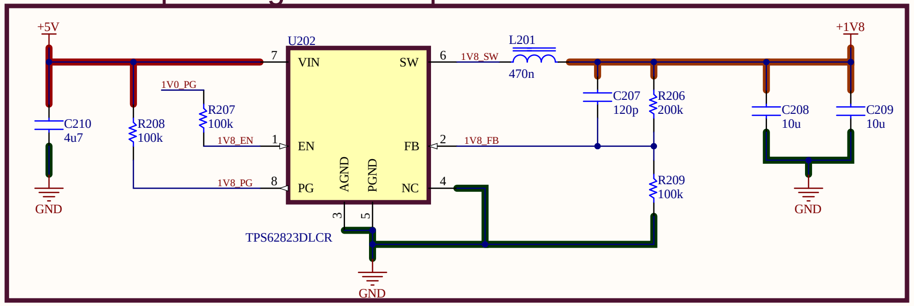
ZHSD POWER SYSTEM

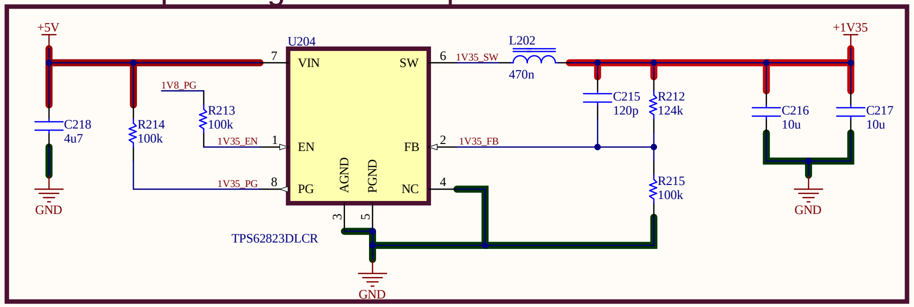
Power Sequencing - 1V Output



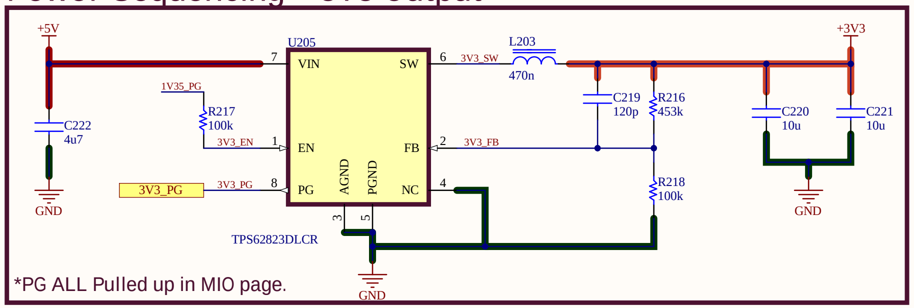
Power Sequencing - 1V8 Output



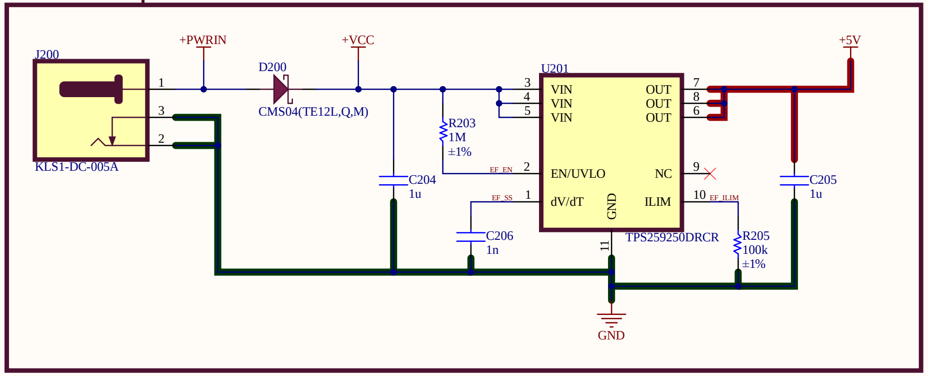
Power Sequencing - 1V35 Output



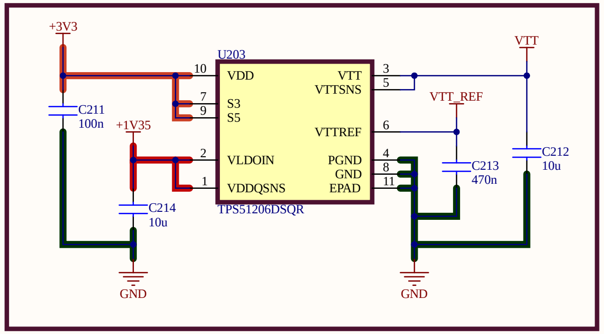
Power Sequencing - 3V3 Output



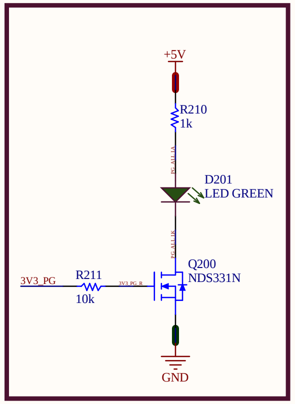
Power Input & EFuse



VTT



PG Led



Power Sequence

*Zynq-7000 SoC DC and AC Switching Characteristics DS187

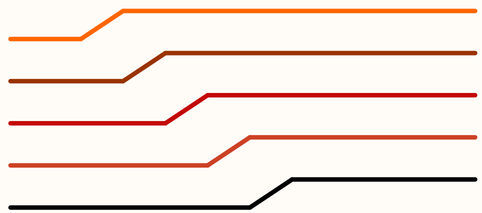
+1V0 VCCPINT, VCCINT, VCCBRAM

+1V8 VCCPAUX, VCCPLL, VCCO_MIO1, VCCAUX

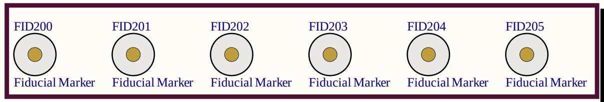
+1V35 VCCO_DDR

+3V3 VCCO_MIO0, VCCO(PL Banks)

+VTT DDR3L Reference Voltage



Fiducials



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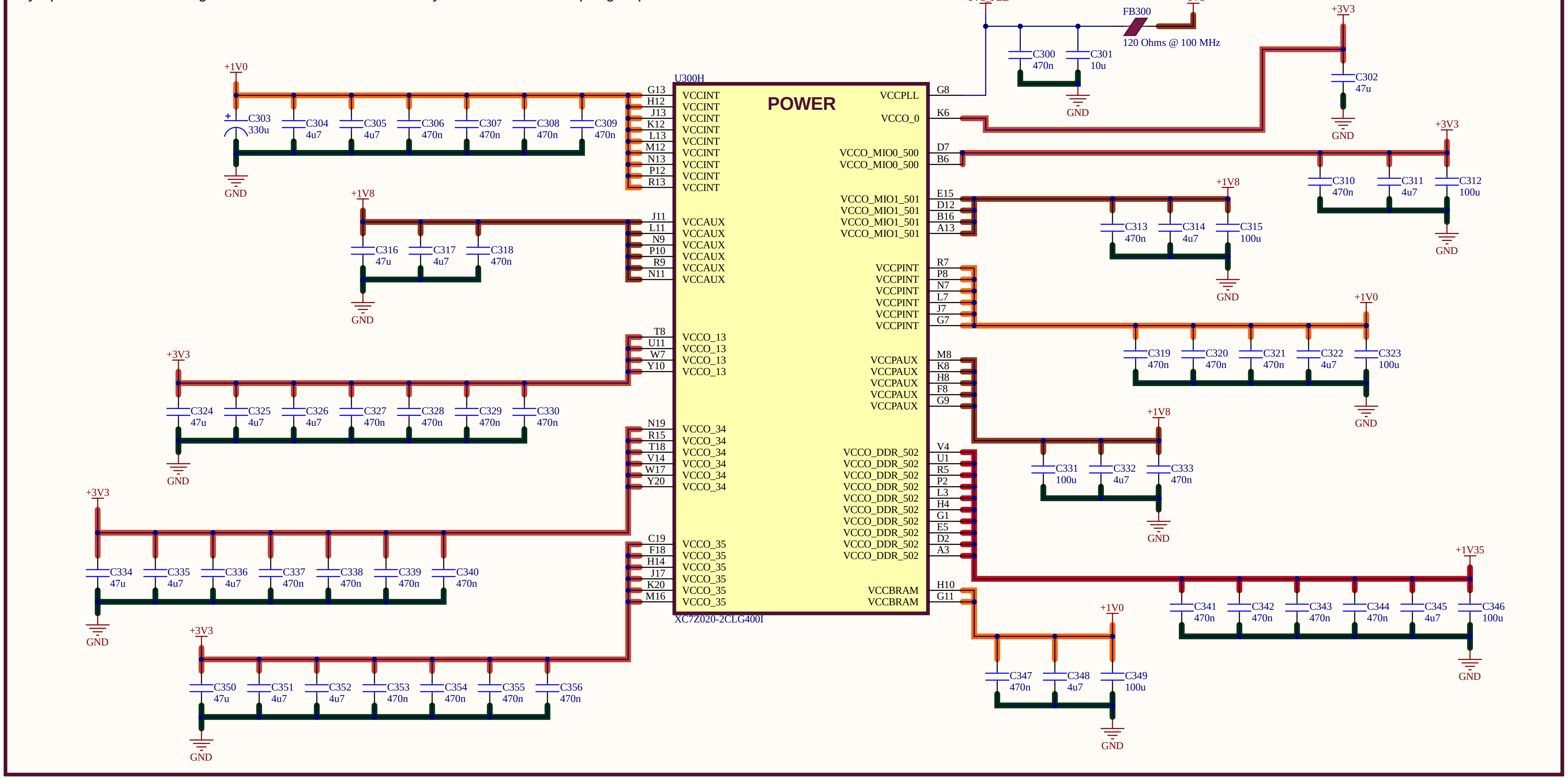
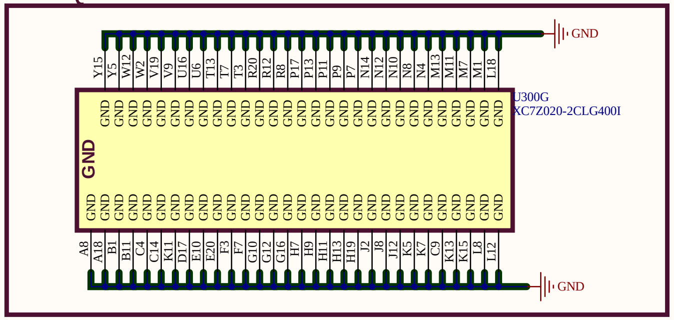
ZHSD ZYNQ POWER DECOUPLING

ZYNQ 7000 Power Decoupling

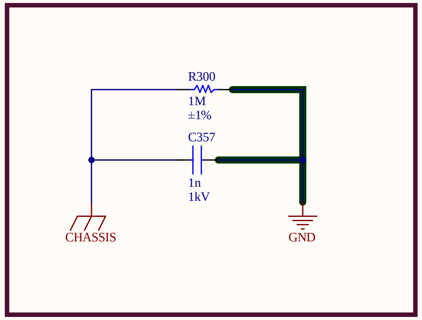
*Zynq-7000 SoC PCB Design Guide -> Power Distribution System -> PCB Decoupling Capacitors



ZYNQ 7000 - GND



ZHSD CHASSIS-GND



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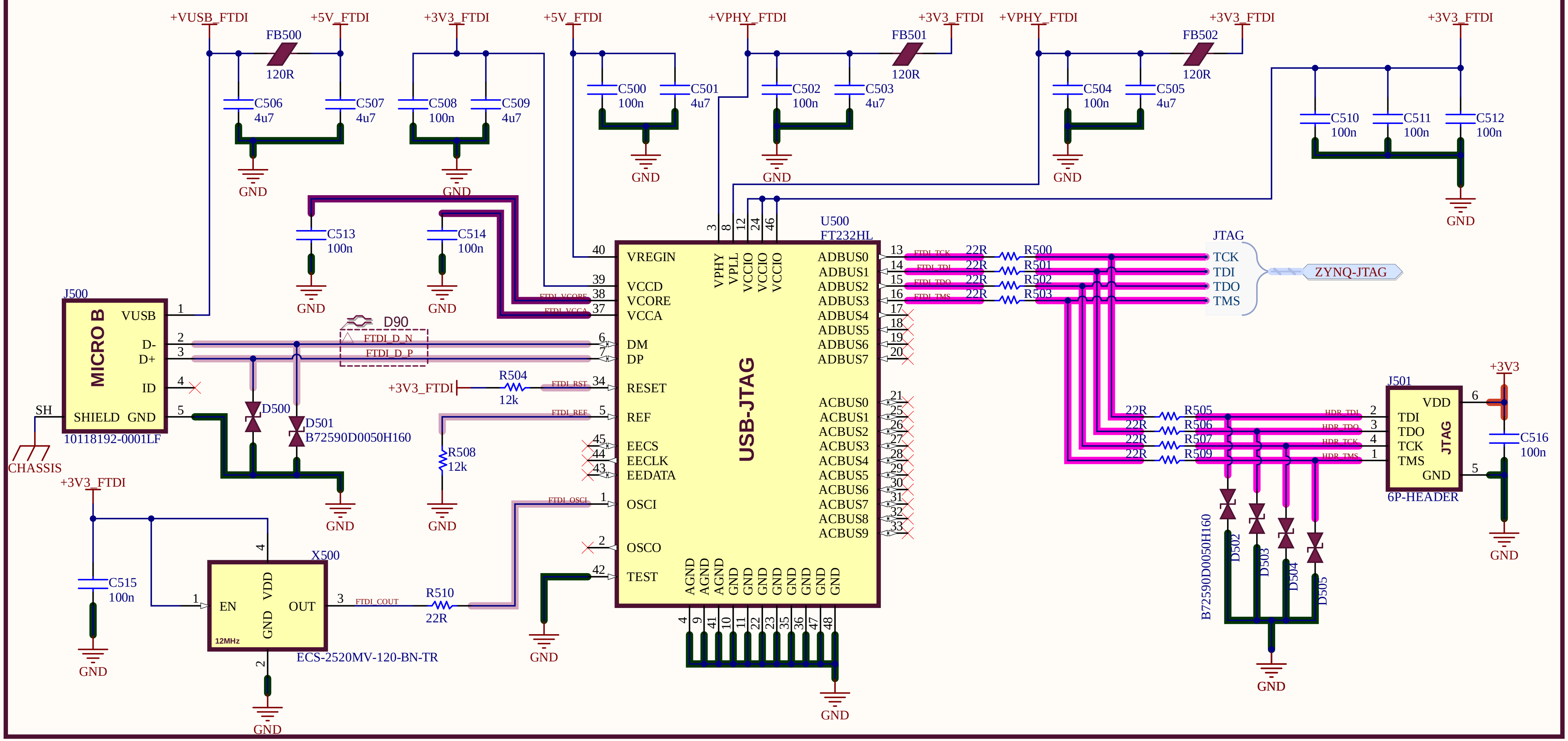
USB-JTAG INTERFACE

USB-JTAG

*Programming and Debugging UG908 -> Appendix E for devices



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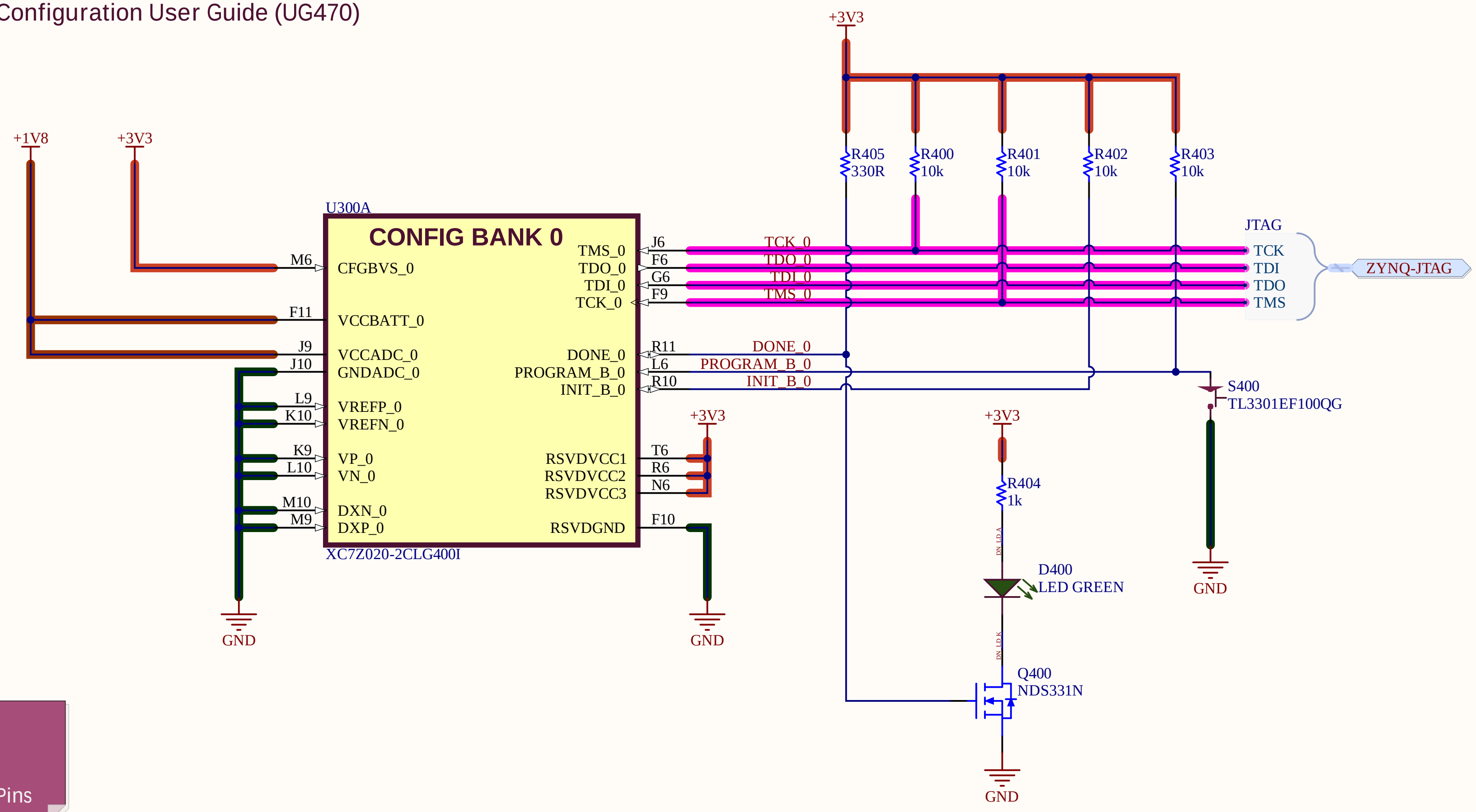
ZHSD ZYNQ CONFIG

ZYNQ-7000 Config Bank 0

*7 Series FPGAs Configuration User Guide (UG470)



UG480 p9 xADC
UG933 p68 DXP/N
UG865 p13 RSVD
UG470 p20 Other Pins

Project Title	ZYNQ-HS-DEV.PrjPcb
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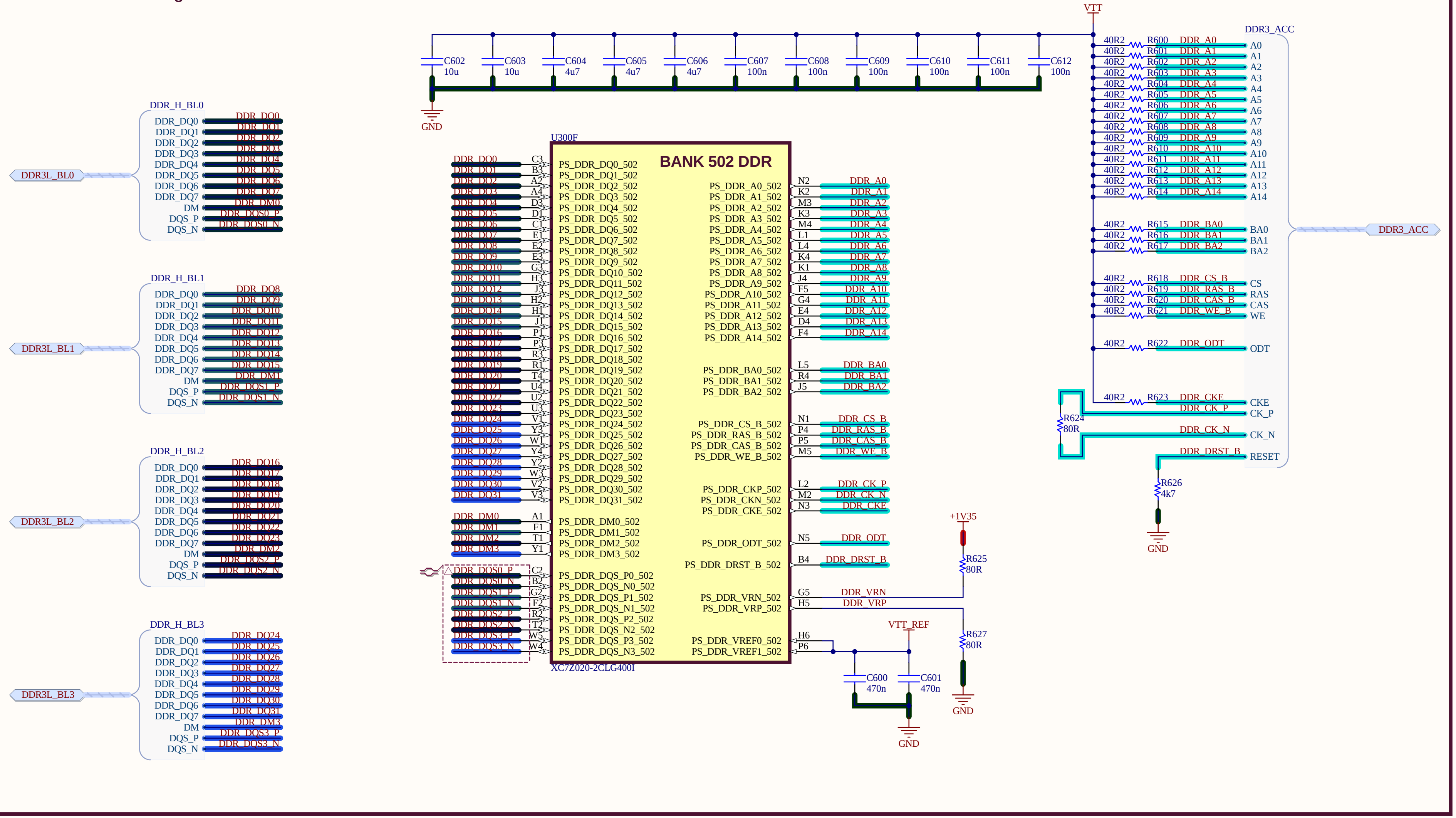
ZHSD ZYNQ DDR3L

ZYNQ-7000 DDR BANK 502

*Zynq-7000 SoC PCB Design Guide UG933



Project Title	ZYNQ-HS-DEV.PrjPcb	
Drawn By Uğur KELEŞ	Sheet Name ZHSD-ZYNQ-DDR.SchDoc	
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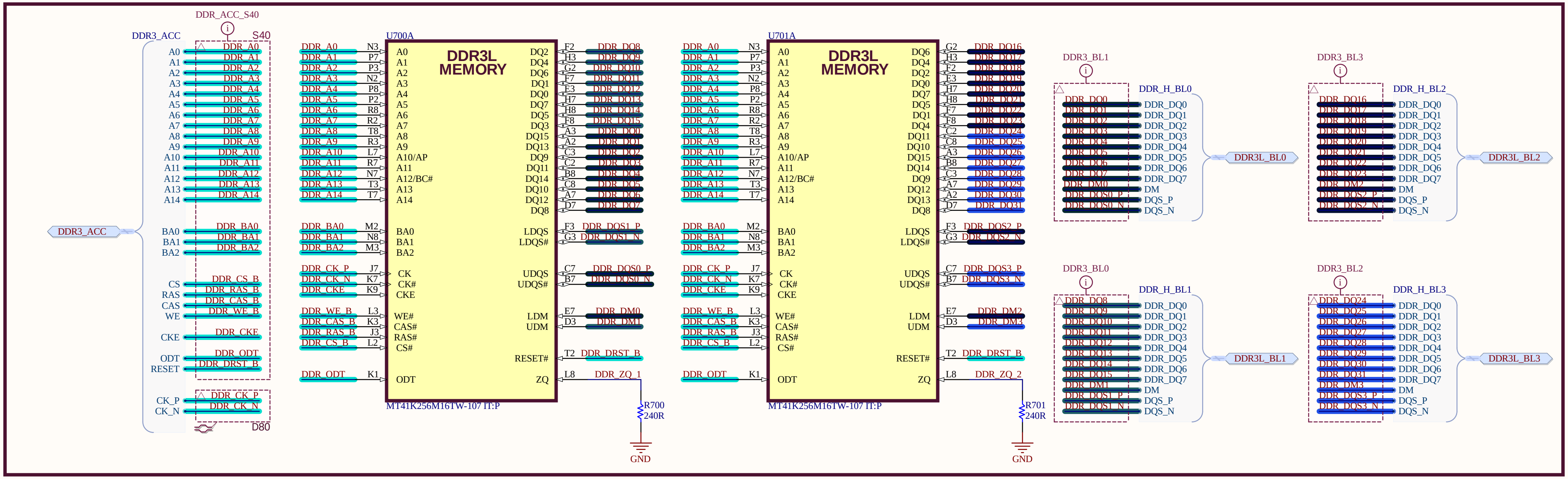
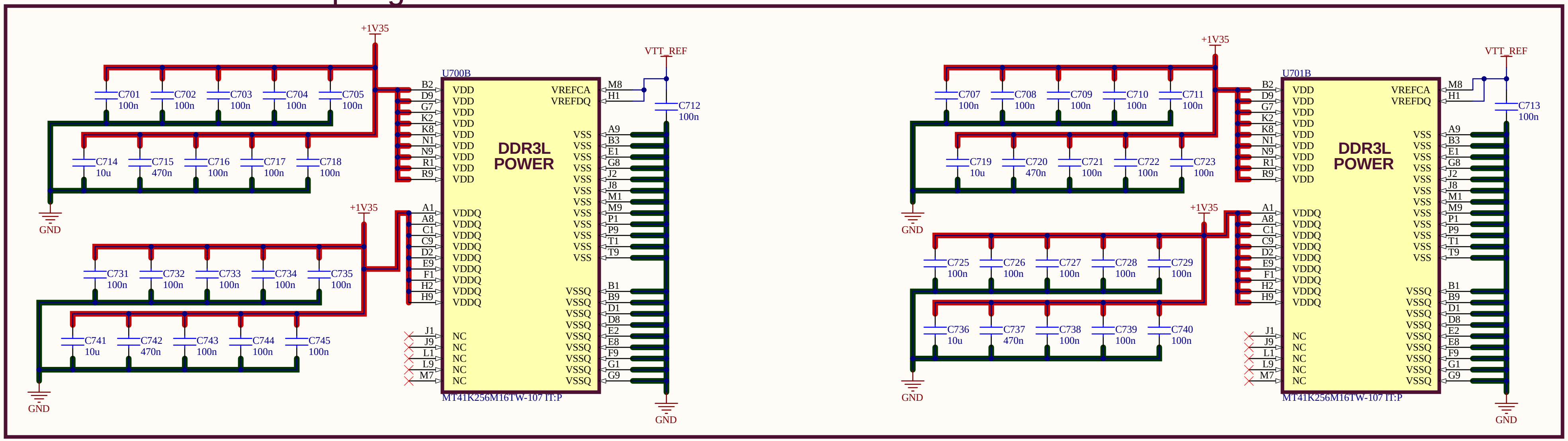
ZHSD DDR3L 1GB

DDR3L IC



DDR3 IC Power Decoupling



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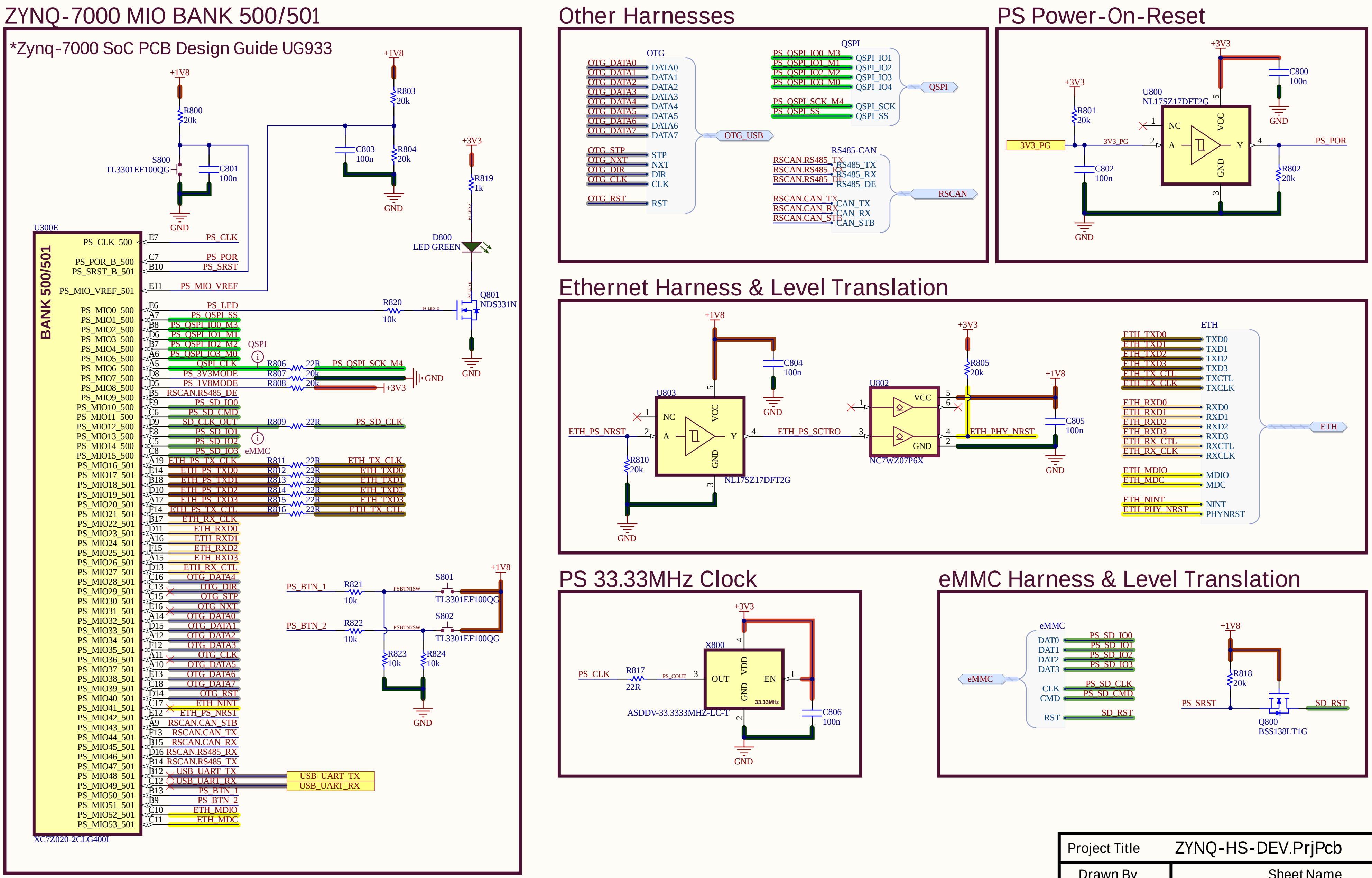
ZHSD ZYNQ PS MIO



Project Title	ZYNQ-HS-DEV.PrjPcb	
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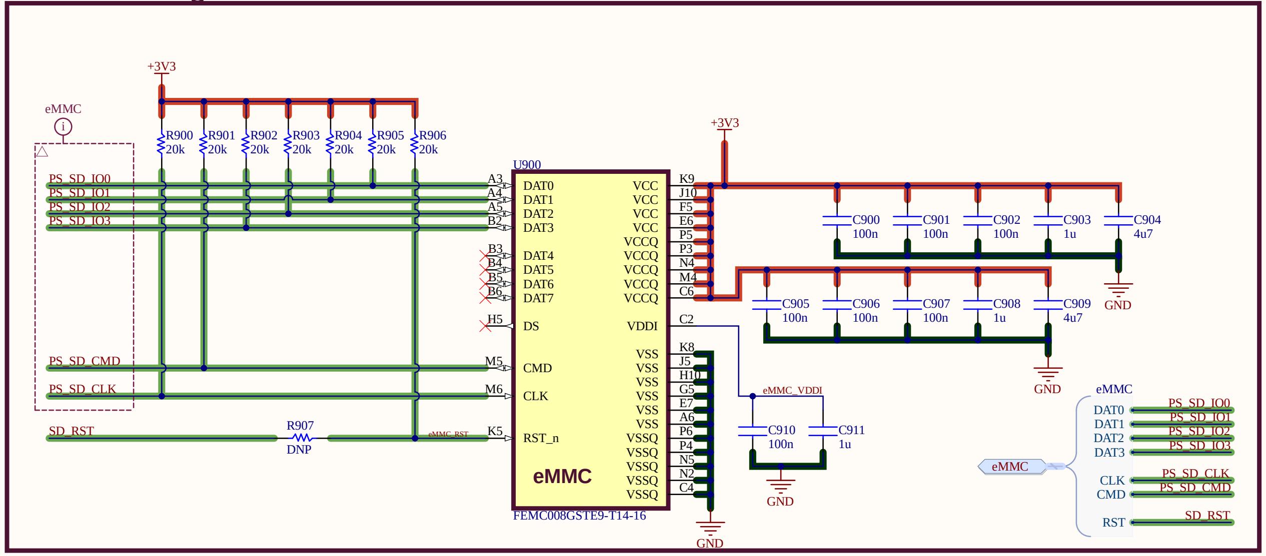
Date 8.02.2026	Sheet 8 / 17	Sheet Size A3

ZHSD BOOTING & STORAGE

eMMC Storage



ZYNQ Boot Modes Reference

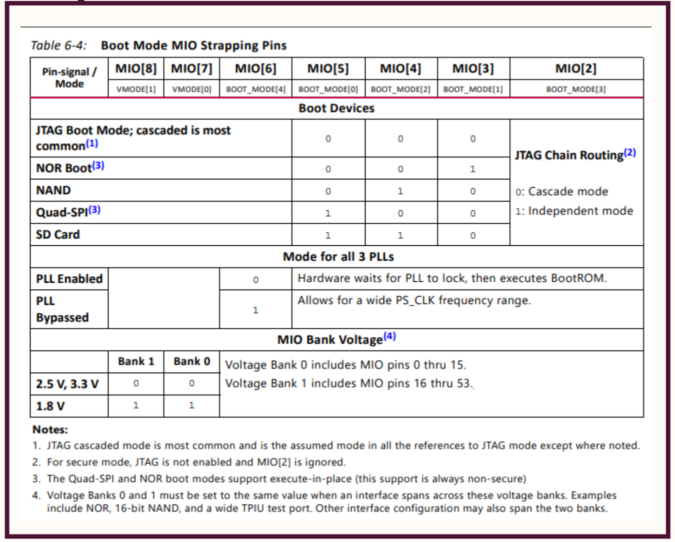


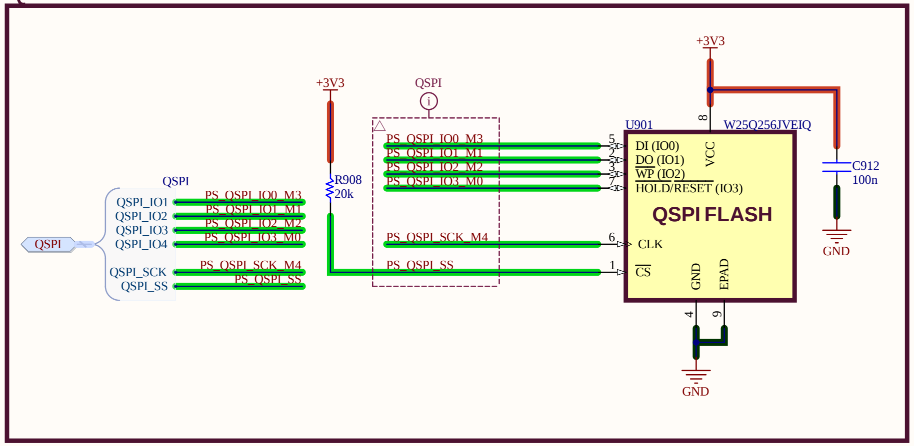
Table 6-4: **Boot Mode MIO Strapping Pins**

Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common(1)				0	0	0	**JTAG Chain Routing**(2) 0: Cascade mode 1: Independent mode
NOR Boot(3)				0	0	1	
NAND				0	1	0	
Quad-SPI(3)				1	0	0	
SD Card				1	1	0	
Mode for all 3 PLLs							
PLL Enabled			0	Hardware waits for PLL to lock, then executes BootROM.			
PLL Bypassed			1	Allows for a wide PS_CLK frequency range.			
MIO Bank Voltage(4)							
	Bank 1	**Bank 0**	Voltage Bank 0 includes MIO pins 0 thru 15. Voltage Bank 1 includes MIO pins 16 thru 53.				
2.5 V, 3.3 V	0	0					
1.8 V	1	1					

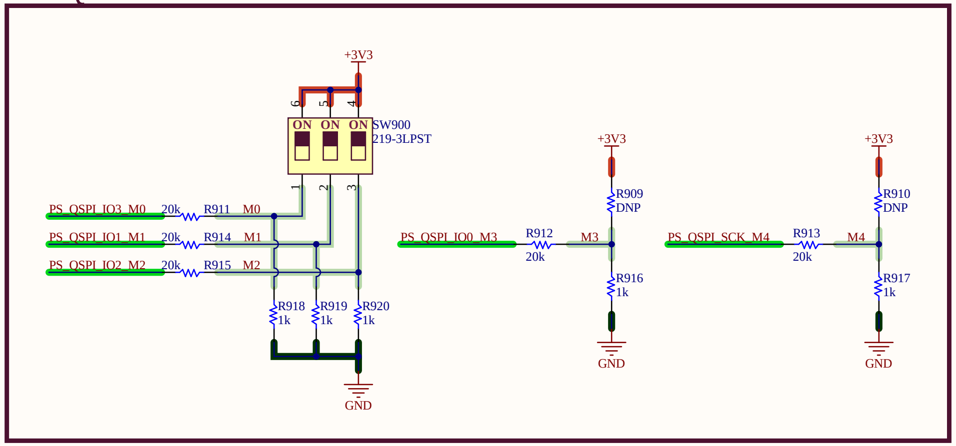
Notes:

1. JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
2. For secure mode, JTAG is not enabled and MIO[2] is ignored.
3. The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure)
4. Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

QSPI Flash



ZYNQ Boot Modes



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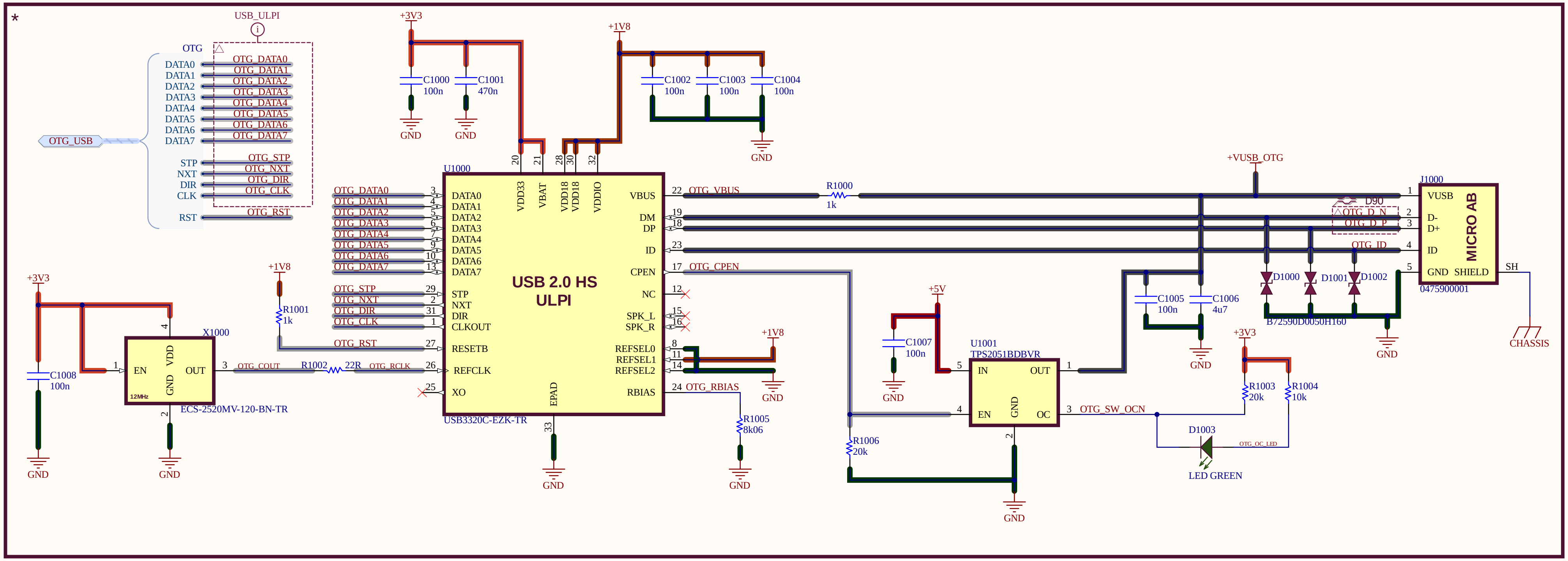
ZHSD USB OTG 2.0 ULPI

USB OTG PHY & Micro AB Connector



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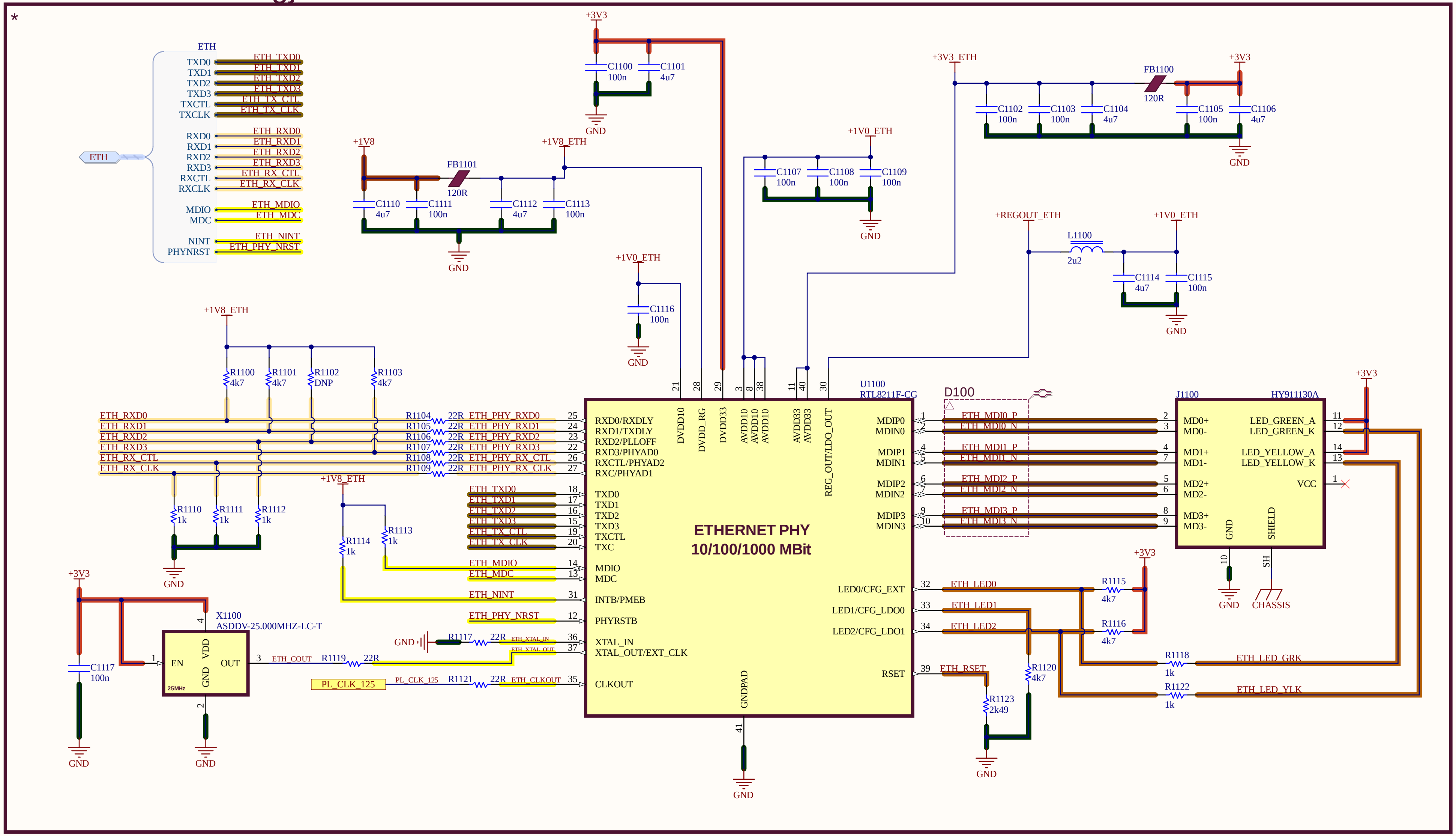
ZHSD GIGABIT ETHERNET

Ethernet PHY & Magjack



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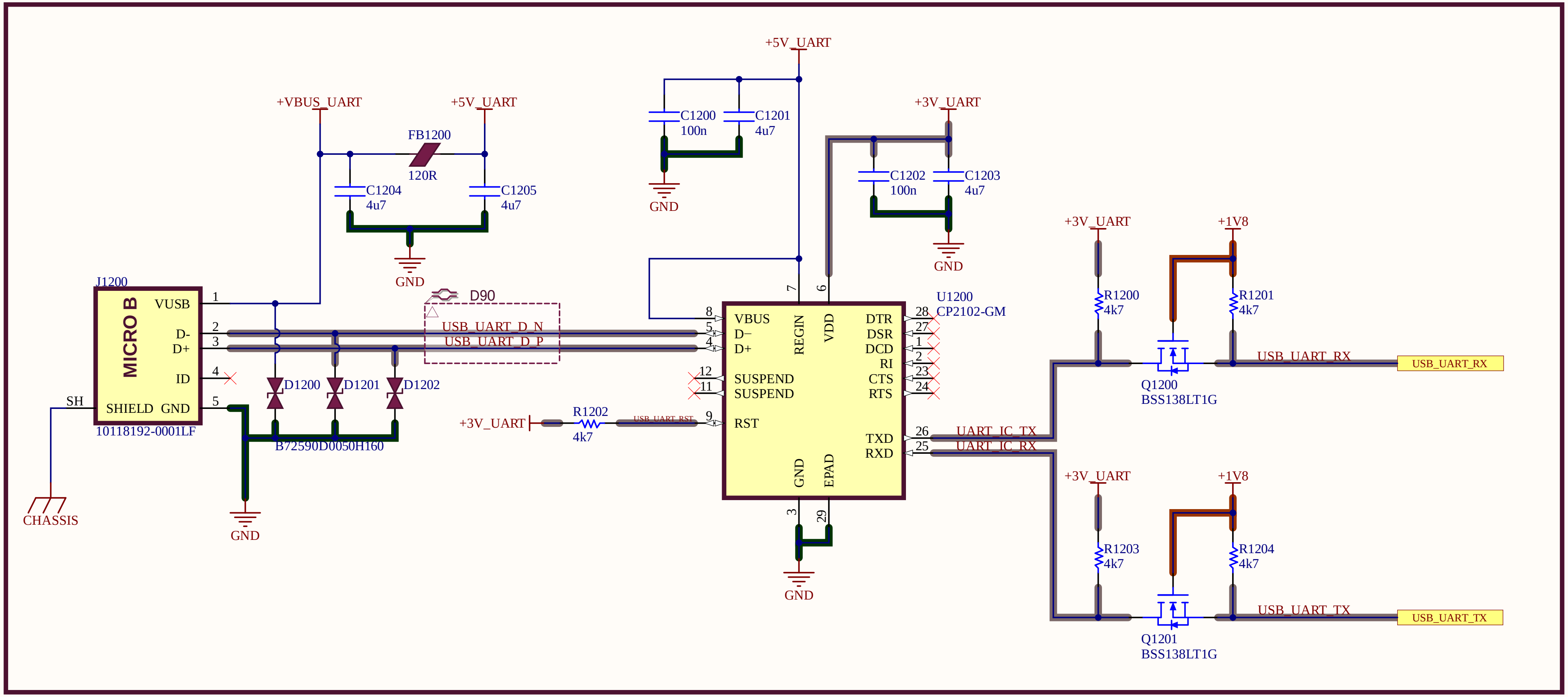
USB-UART INTERFACE

USB-UART



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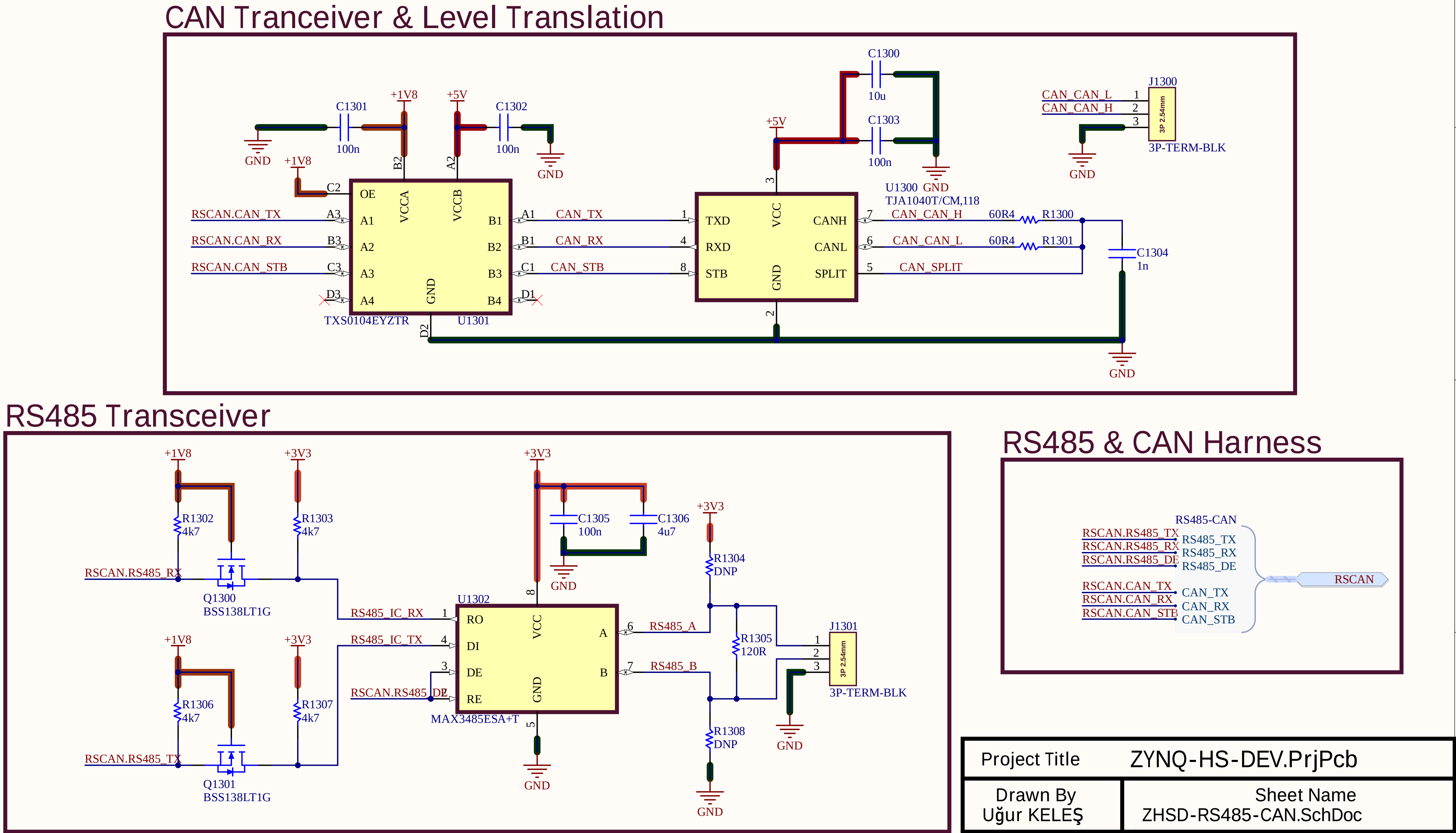
RS485 & CAN INTERFACES



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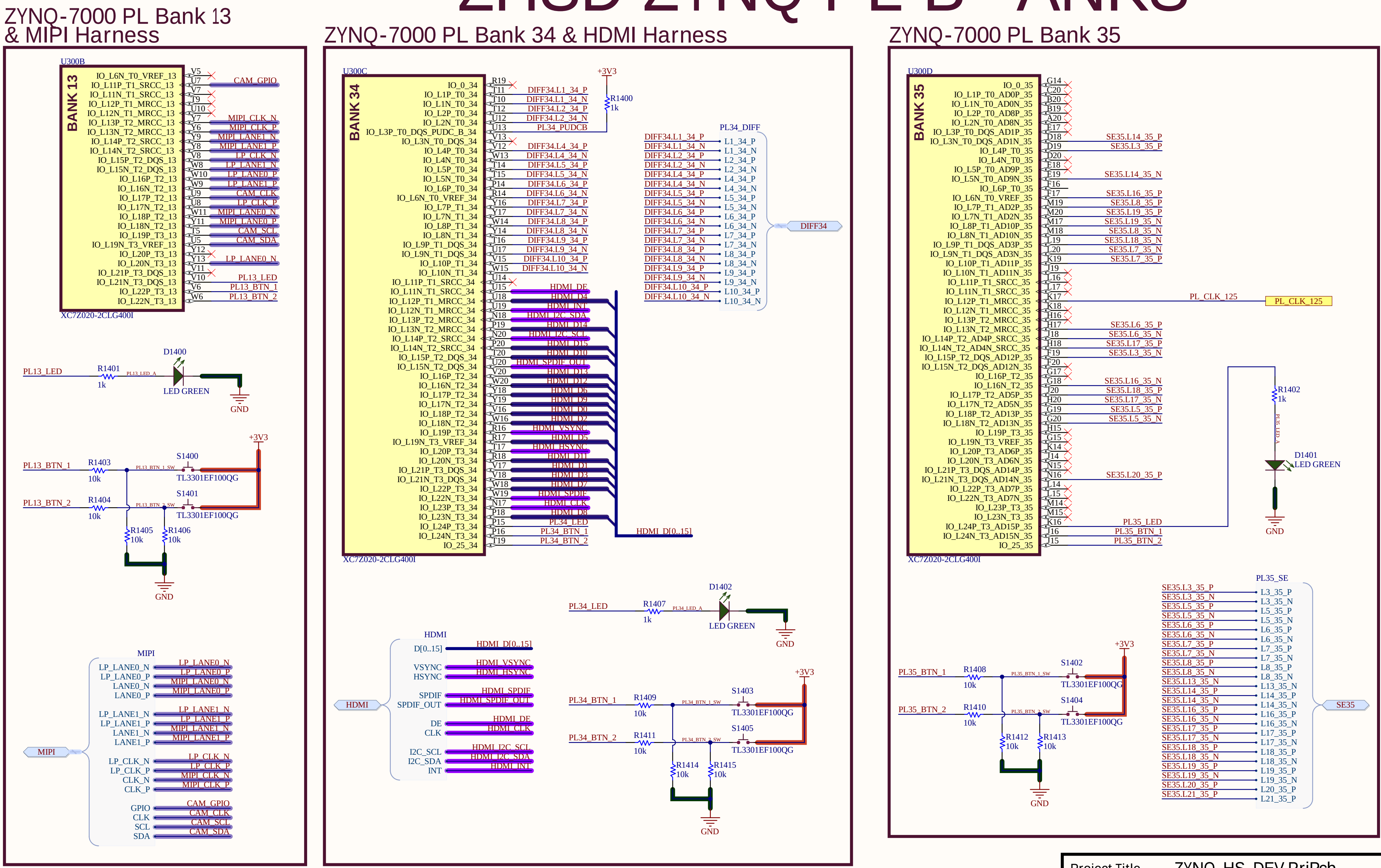
Date 8.02.2026	Sheet 13 / 17	Sheet Size A4	

ZHSD ZYNQ PL B ANKS



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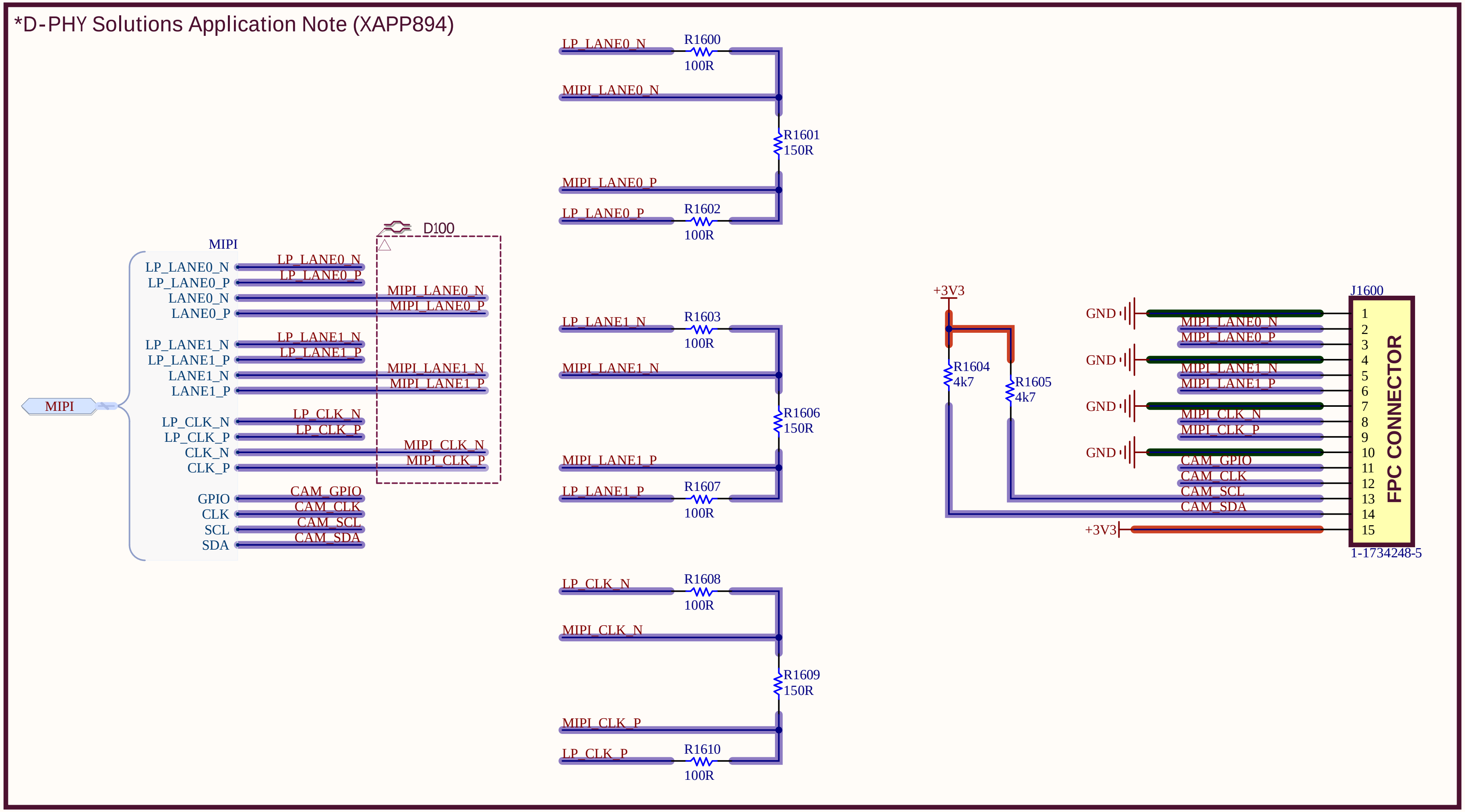
ZHSD MIPI INTERFACE

MIPI CSI-2



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Date 7.02.2026	Sheet 15 / 17
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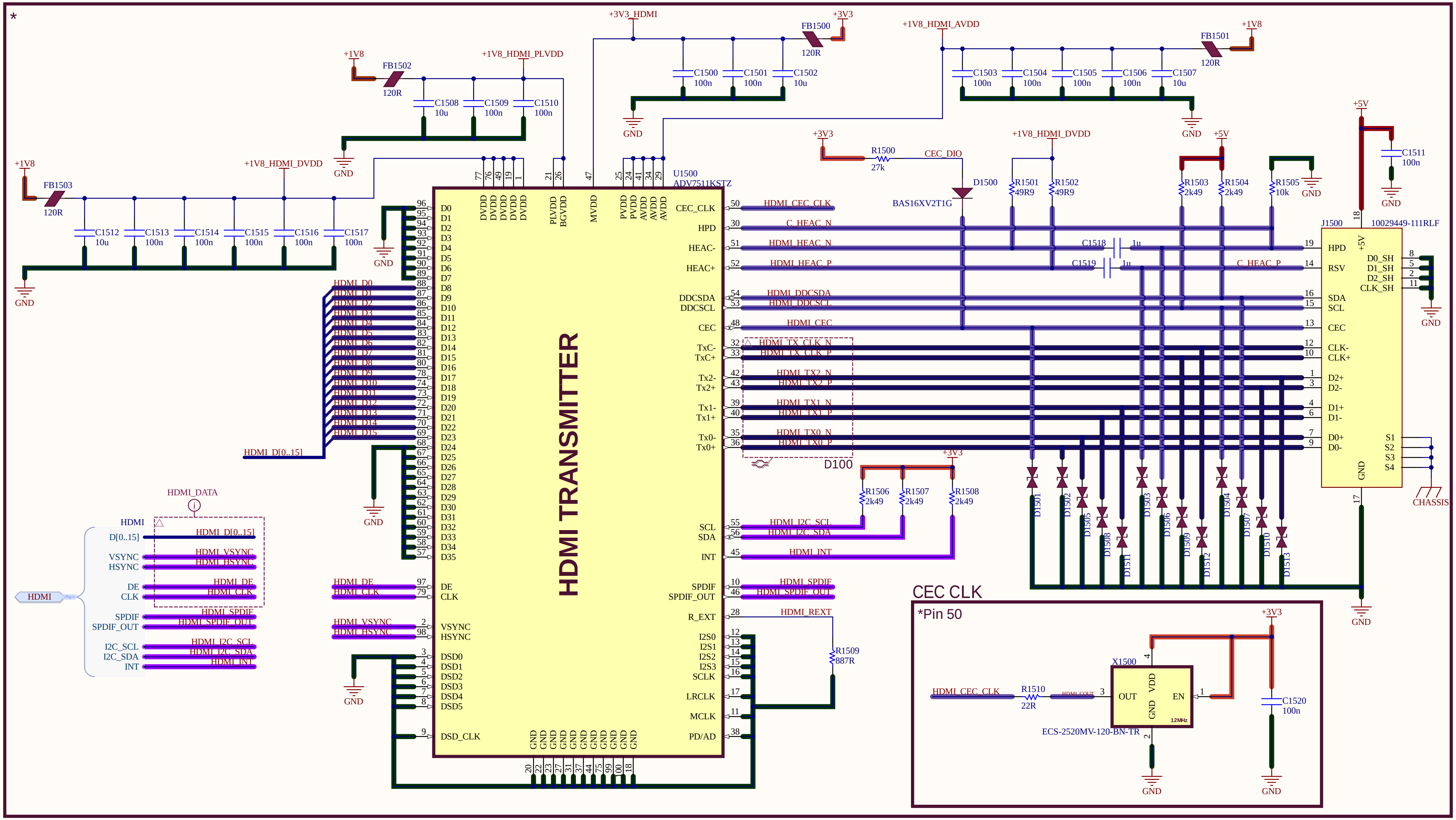
ZHSD HDMI INTERFACE

HDMI Transmitter and Connector



Project Title	ZYNQ-HS-DEV.PrjPcb	
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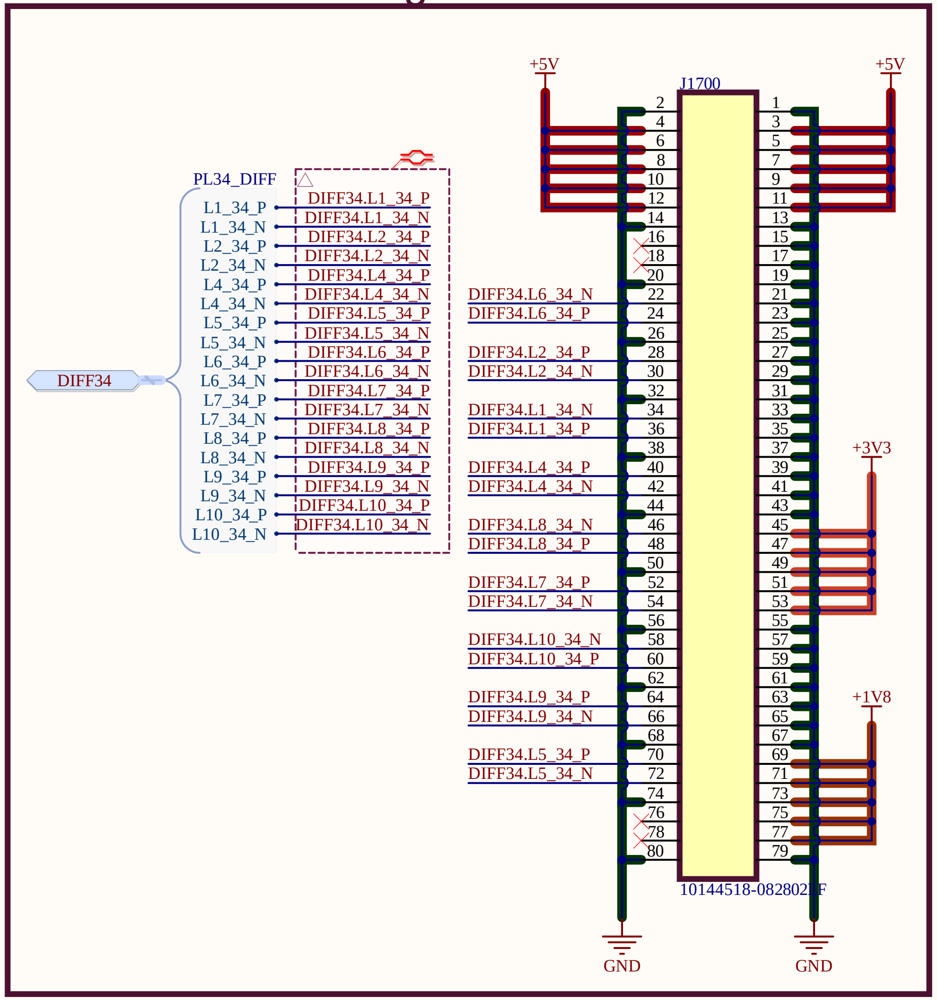
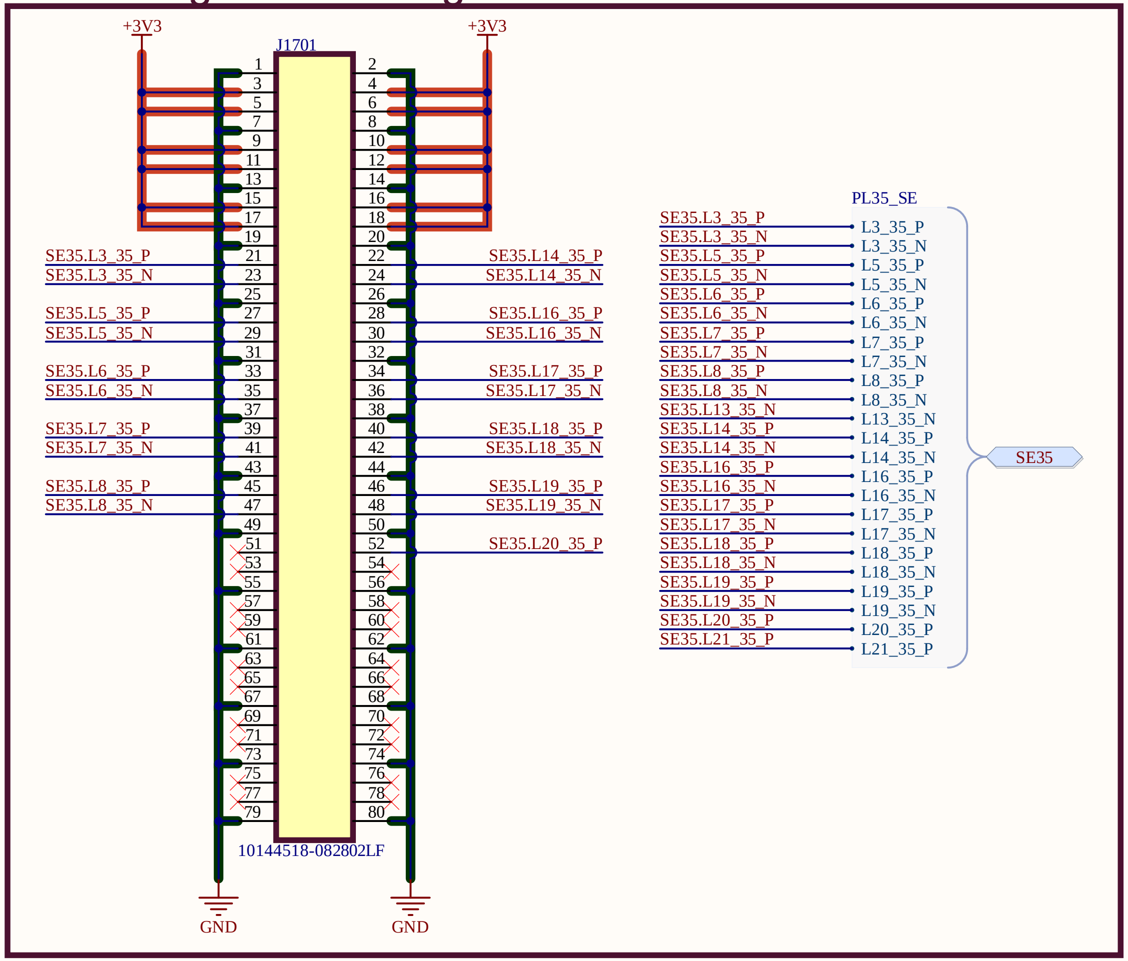
Date 8.02.2026	Sheet 15 / 17	Sheet Size A3

ZHSD PL CONNECTIONS

PL34 Differential Signals



PL35 Single Ended Signals



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